Computational triage of therapeutic and germline vulnerabilities in IDH–wildtype glioblastoma: a focal–adhesion dependency in CDKN2A/B–deleted tumors

Brad Moore*

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Dedicated to James E. Moore (1961–2025).

Abstract

Glioblastoma (GBM) remains almost uniformly lethal, and the CDKN2A/B–homozygous–deleted, IDH–wildtype subtype carries a particularly poor prognosis with no subtype–specific therapy. Using publicly available functional–genomic, transcriptomic, pharmacologic, clinical, and population–genetic datasets, we performed an integrated triage for actionable vulnerabilities in this subtype, and report three findings of differing strength. **(1)** A focal–adhesion/FAK dependency in CDKN2A/B–null GBM is supported by *five complementary lines of evidence* (Figure 1): it is selectively essential by CRISPR knockout (*ITGAV* $q = 1.8 \times 10^{-4}$, *TLN1*, *FERMT2*, *ITGB5*, *PTK2*/FAK, *ILK*); it reproduces under *orthogonal RNAi* silencing (*TLN1* $p = 3.7 \times 10^{-6}$, FAK $p = 1.3 \times 10^{-3}$, *ITGAV* $p = 5 \times 10^{-3}$), ruling out a copy–number cutting artifact; a *potent* FAK inhibitor (PF–562271, GDSC dose–response) selectively sensitizes CDKN2A/B–null lines ($p = 2 \times 10^{-3}$), closing the gap left by the inert first–generation probe (GSK2256098) in PRISM; *ITGAV* mRNA is upregulated in CDKN2A/B–null TCGA–GBM tumors ($q = 0.014$); and FAK is both more abundant and more *activated* (autophosphorylation–Y397 $p = 0.012$; FAK protein $p = 3 \times 10^{-4}$; *FERMT2* protein $p = 9 \times 10^{-5}$) in CDKN2A/B–null CPTAC–GBM tumors. The signal is not a mesenchymal–subtype artifact (null tumors are not mesenchymal–enriched, $p = 0.40$). One major caveat constrains attribution: CDKN2A/B is co–deleted with the adjacent 9p21 gene *MTAP* in ∼92% of null lines, and the two deletions are statistically inseparable in DepMap (our positive control, the canonical *MTAP*–loss dependency *PRMT5*, is itself misattributed to CDKN2A/B), so the dependency is most defensibly described as *9p21–codeletion–associated*—though it does not vary with *MTAP* status among null lines, and CDKN2A/B is the more mechanistically plausible driver. It remains a genetic/protein–level dependency awaiting wet–lab pharmacologic confirmation. **(2)** A large, molecularly–definable CDK4/6–inhibitor–eligible subset (IDH–wildtype, RB1–intact, CDK4/6–axis–activated; ∼65% of the full GBM cohort, ∼70% of IDH–wildtype GBM) has never been prospectively enrolled with resistance stratification; eligibility conferred no survival advantage in TCGA or MSK–IMPACT cohorts, and a large fraction of eligible tumors carry PTEN/PI3K–axis alterations (PTEN mutation 34%, deletion 12%, PIK3R1/PIK3CA ∼10% each) that plausibly explain prior null trials.

*Independent researcher. Correspondence: moore.brad.m@gmail.com.

(3) A Finnish–founder *MUTYH* pathogenic allele (p.Gly155Asp, rs587781864) shows a suggestive monoallelic association with GBM in FinnGen R13 (OR= 24.9, ∼4 case carriers) that is internally validated by expected colorectal signals but does not survive multiple–testing correction. Together these define one strong genetic dependency requiring better pharmacologic tools, one trial–design hypothesis with an explicit resistance map, and one replication–grade germline lead.

1 Introduction

IDH–wildtype glioblastoma is the most common and most aggressive primary brain tumor, with median overall survival of ∼15 months despite maximal therapy. Homozygous deletion of the CDKN2A/CDKN2B locus on chromosome 9p21 occurs in $> 55\%$ of cases and marks a worse–prognosis group, yet no therapy specifically exploits this lesion: the deletion removes tumor–suppressor function (no direct drug target) and, while it sensitizes in principle to CDK4/6 inhibition, clinical results have been disappointing.

We took an integrative, hypothesis–generating approach across complementary public datasets—functional dependency (DepMap CRISPR), tumor expression (TCGA–GBM RNA–seq), drug response (PRISM), somatic landscape and survival (TCGA, MSK–IMPACT), and germline population genetics (FinnGen R13, gnomAD)—to triage candidate vulnerabilities in the CDKN2A/B–deleted, IDH–wildtype subtype. Our aim is explicitly to separate findings that are robust from those that are suggestive, and to state clearly where the evidence fails.

2 Results

2.1 A focal–adhesion / FAK–integrin dependency in CDKN2A/B–null GBM (Figure 1)

We classified 134 CNS cancer cell lines by CDKN2A/B copy number (relative CN<0.2 = homozygous deletion; 55 null, 79 intact) and compared gene–level CRISPR dependency (Chronos effect) between groups. The genes most selectively essential in null lines form a coherent **integrin → talin/vinculin/kindlin → FAK → actomyosin/Arp2/3** module rather than scattered hits (Figure 1A): *ITGAV* ($\Delta = -0.398$, $q = 1.8 \times 10^{-4}$), *TLN1* ($q = 8.8 \times 10^{-5}$), *VCL* ($q = 7.4 \times 10^{-5}$), *FERMT2* ($q = 0.020$), *ITGB5* ($q = 2.5 \times 10^{-3}$), *PTK2*/FAK ($q = 0.037$), *ILK* ($q = 0.016$), plus *PPP1R12A*, *RAC1*, the Arp2/3 complex (*ARPC2/3/4*, *ACTR2/3*), and downstream *JUN*/*FOSL1* (AP–1) and *WWTR1*/*TEAD1* (Hippo/YAP). The directionality is uniform: every member is *more* essential in CDKN2A/B–null lines.

Orthogonal RNAi silencing (Figure 1B). Because CRISPR knockouts can produce copy–number–related cutting toxicity, we re–tested the module in the DepMap RNAi (DEMETER2 combined) dataset, which silences by shRNA and does not cut DNA. The core focal–adhesion module reproduced in CDKN2A/B–null lines: *TLN1* ($p = 3.7 \times 10^{-6}$), *FERMT2* ($p = 7.9 \times 10^{-4}$), *PTK2*/FAK ($p = 1.3 \times 10^{-3}$), *ITGAV* ($p = 5.0 \times 10^{-3}$), *ILK* ($p = 5.1 \times 10^{-3}$) and *ITGB5* ($p = 0.048$); 6 of 7 core genes replicated (only *VCL* did not), and the three anchors (*ITGAV*, *TLN1*, FAK) remained one–sided $p < 0.05$ even in the small CNS subset. Two independent silencing technologies converging excludes a CRISPR–specific artifact.

Tumor transcriptome and proteome. In TCGA–GBM RNA–seq, *ITGAV* was upregulated in null tumors ($p = 0.0023$, $q = 0.014$; Figure 1F), and the expected cell–cycle rewiring was confirmed in the same comparison (*RB1*↑, *CDK6*↑, *E2F1*↓; each $p < 10^{-4}$), validating the classification. Moving closer to function, in CPTAC–GBM proteogenomics (44 deep CDKN2A/B–deleted vs 36 intact tumors) **FAK was both more abundant and more activated** in null tumors: FAK autophosphorylation at **Y397**—the canonical activation site—was elevated (null>intact $p = 0.012$; Figure 1D), as were FAK total protein ($p = 3 \times 10^{-4}$) and *FERMT2* protein ($p = 9 \times 10^{-5}$) and phospho ($p = 1 \times 10^{-4}$; Figure 1E). The signal was specific to FAK–activation and *FERMT2* rather than a blanket adhesion–protein increase.

A potent FAK inhibitor closes the pharmacologic gap (Figure 1C). The only FAK inhibitor in the PRISM repurposing screen, GSK2256098, had shown no selective killing of null lines—but it was essentially inert across all CNS lines at the screened dose (2.5 µM), an underpowered test rather than a refutation. We therefore turned to GDSC, which provides proper dose–response for the *potent* FAK inhibitor PF–562271. **CDKN2A/B–null lines were selectively more sensitive** (AUC null<intact, pan–cancer $p = 2.0 \times 10^{-3}$; $\ln\mathrm{IC}_{50}$ $p = 0.022$), with the same direction in the CNS subset (underpowered, $n = 29/25$). The effect size is modest ($\sim$2% AUC shift), consistent with FAK inhibition being partly cytostatic in 2D viability and arguing for clonogenic/3D readouts in wet–lab validation.

Not a mesenchymal–subtype artifact. The focal–adhesion program is biologically coupled to the mesenchymal GBM state (FAK–module vs mesenchymal signature Spearman $\rho = 0.72$, $p = 9 \times 10^{-14}$). However, CDKN2A/B–null tumors were *not* mesenchymal–enriched (signature null vs intact $p = 0.40$), and the null–specific signal manifests at the protein/activation and dependency levels rather than as bulk transcription—i.e. a functional vulnerability, not a transcriptional–subtype label. Single–cell analysis of the Neftel atlas (GSE131928; 31–33 patients, malignant cells only after stromal exclusion) confirms this coupling is intrinsic to malignant cells rather than a bulk–mixing artifact: tumor–level mean FAK–module expression tracks mesenchymal content strongly (Smart–seq2 $r = 0.81$, $p = 4 \times 10^{-8}$), though the relationship is a tumor–ecosystem property (patient–level $r^2 \approx 0.65$) far more than a cell–intrinsic one (per–malignant–cell $r^2 \approx 0.13$), with no residual FAK structure across Neftel cell states after mesenchymal adjustment. Because this near–uniformly CDKN2A/B–deleted cohort lacks an intact comparator, single–cell data cannot test null–specificity—which therefore continues to rest on the CPTAC null–vs–intact contrast above, not on tumor expression.

The CDKN2A/B vs *MTAP* (9p21 co–deletion) confound. *MTAP* lies immediately telomeric to CDKN2A on 9p21 and is co–deleted in 92% of CDKN2A/B–null DepMap lines (indicator collinearity $\phi = 0.92$); only 30 of 354 null lines pan–cancer (4 of 54 in CNS) carry CDKN2A/B loss with *MTAP* retained. Because the canonical 9p21–deletion collateral dependency is *MTAP*–loss→*PRMT5*/*MAT2A* (not FAK), we tested whether the FAK signal is separable from *MTAP* loss. Two results bound the answer. First, reassuringly, FAK–module dependency does *not* differ between *MTAP*–co–deleted and *MTAP*–intact null lines (all seven genes $p > 0.09$), arguing against an *MTAP*–driven effect; and in the cleanest contrast (*MTAP*–intact null vs intact) all seven module genes remain direction–consistent (more essential in null), albeit underpowered and individually non–significant ($n = 30$; best *TLN1* $p = 0.07$). Second, a joint regression cannot adjudicate, because the two deletions are so collinear that even the positive control misassigns: effect $\sim$ CDKN2A/B–null + *MTAP*–null attributes the *known* *MTAP* dependency *PRMT5* to CDKN2A/B ($p = 3 \times 10^{-4}$) rather than *MTAP* ($p = 0.65$). We therefore tested the subset contrast directly with a permutation/bootstrap meta–analysis pooling CRISPR and RNAi (50,000 label permutations per gene; screens combined by Stouffer's Z). Here the focal–adhesion module shows a coherent residual dependency in *MTAP*–intact null lines—*PTK2* (pooled $p = 0.017$), *FERMT2* (0.014) and

PXN (0.038) reaching nominal significance, *ITGAV*/*TLN1*/*WWTR1* trending ($p = 0.07$–0.08)—while the *MTAP*–loss positive controls collapse in the very same contrast (*PRMT5* from pooled $p \approx 5 \times 10^{-9}$ in the confounded comparison to 0.054; *MAT2A* to 0.41). This dissociation—*MTAP*–driven controls falling away when *MTAP* is held intact, while the adhesion module persists—is the strongest cell–line evidence that the FAK signal is not merely *MTAP*–collateral. Its limits are explicit: no single module gene survives multiple–testing correction in this contrast, the effect is sensitive to the deletion call (it dissolves under a strict homozygous–both definition, $n = 6$), and the separation is statistical, not experimental. On mechanism the dependency is more plausibly a consequence of CDKN2A/B–CDK4/6–RB rewiring (adhesion/mechanotransduction) than of *MTAP*/methionine–salvage metabolism, but isogenic models are required to settle attribution.

Summary: five complementary technologies—CRISPR, orthogonal RNAi, potent–inhibitor dose–response, tumor RNA, and tumor protein/phospho—concordantly support an integrin/FAK focal–adhesion dependency in 9p21–codeleted (CDKN2A/B–null) GBM, now including the protein–level functional readout (phospho–FAK Y397) and a proper pharmacologic tool. The remaining gap is dedicated wet–lab validation of potent, CNS–relevant FAK inhibitors (e.g. defactinib/VS–6063, PF–573228), ideally combined with CDK4/6 inhibition, since adhesion signaling is a recognized route of CDK4/6–inhibitor resistance. We emphasize that this is a dependency hypothesis, not a repurposing recommendation: FAK–inhibitor monotherapy has been clinically unproductive in unselected solid tumors and in GBM, and a prior CDKN2A–stratified FAK trial in *KRAS*–mutant NSCLC was negative (§3)—so the bar for translation is high and combination/selection strategies are required.

2.2 The transcriptional output node: TAZ/TEAD1 is co–essential, 9p21–separable, and druggable (Supplementary Figure S2)

The terminal effector of the integrin/FAK/mechanotransduction axis is the Hippo transcriptional module YAP/TAZ–TEAD. Because FAK has been clinically unproductive in GBM (§3), we asked whether this transcriptional *output* is independently essential and better positioned pharmacologically. In the same genome–wide CNS CRISPR scan (55 null vs 79 intact), *WWTR1*/TAZ ranks among the strongest selective dependencies ($\Delta = -0.225$, $q = 6.2 \times 10^{-4}$, rank 34 of 17,916—more statistically significant than *PTK2*/FAK itself), and *TEAD1* is also selectively essential ($\Delta = -0.180$, $q = 0.026$, rank 68; Supplementary Figure S2A). The signal is paralog–specific: *YAP1* ($q = 0.17$) and *TEAD2/3/4* are not selective—as expected if the redundant *YAP1*$\leftrightarrow$*WWTR1* and *TEAD1–4* families buffer single–gene knockout, so individual–gene essentiality *understates* pathway dependence and a pan–TEAD (not YAP1–selective) strategy is indicated.

Orientation control. If the dependency is genuinely on YAP/TAZ *activity*, the Hippo *negative* regulators that restrain YAP/TAZ (*NF2*, *LATS1/2*, *SAV1*, *MOB1A/B*) should not be essential and may favour growth when removed. They behave exactly so: none is selectively essential, and several are among the most growth–*promoting* knockouts genome–wide in null lines (*NF2* rank 17,870 of 17,916; Supplementary Figure S2B)—the expected functional signature of a YAP/TAZ–dependent state, and an internal control that the axis is read in the correct direction.

9p21/*MTAP* separation. In *MTAP*–intact null lines (CDKN2A/B–null with *MTAP* retained) TAZ is more essential than in intact lines (pan–cancer Mann–Whitney $p = 0.032$, $n = 30$; CNS $p = 0.005$, $n = 4$; Supplementary Figure S2C). The more conservative permutation meta–analysis softens the pan–cancer signal (pooled CRISPR+RNAi $p = 0.068$) while the CNS contrast holds (permutation $p = 0.035$, $n = 4$); under the same analysis several focal–adhesion genes reach nominal

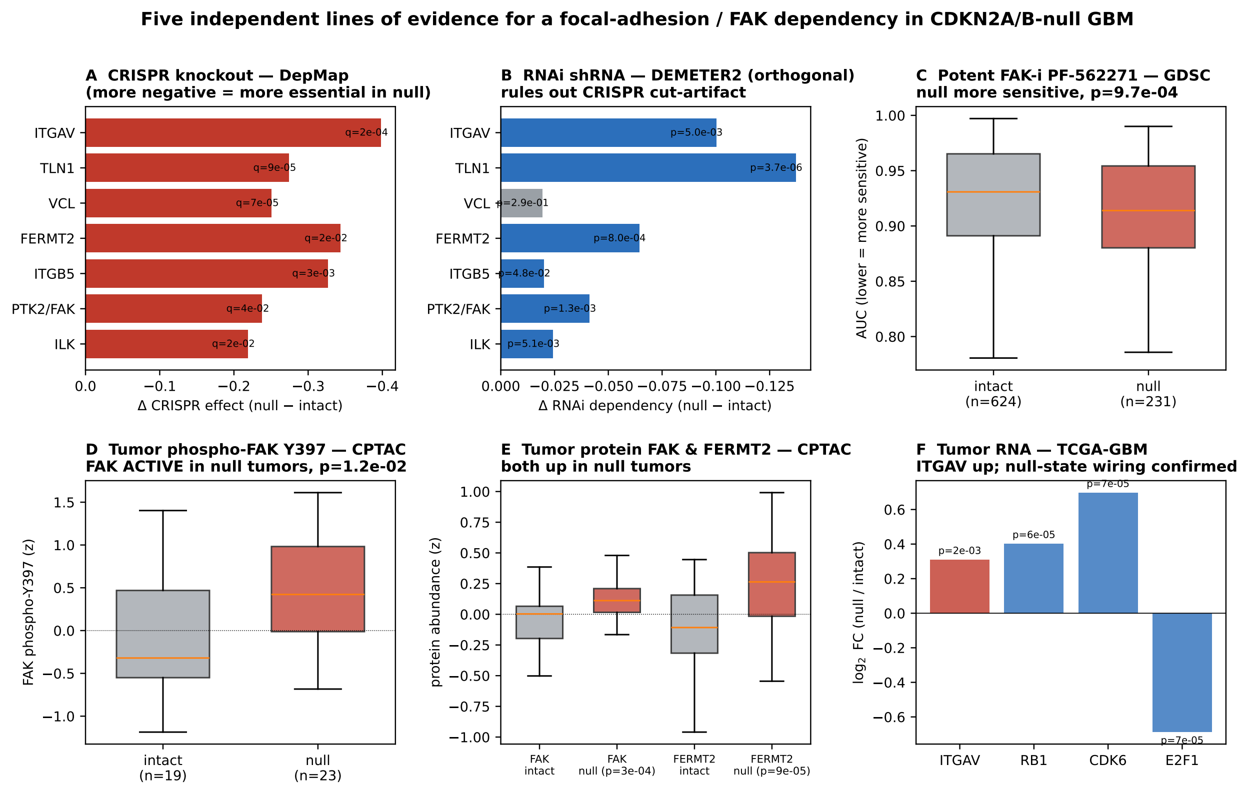

Figure 1: Five complementary lines of evidence for a focal–adhesion / FAK dependency in CDKN2A/B–null glioblastoma. **(A)** DepMap CRISPR knockout: differential essentiality (Δ = mean effect in CDKN2A/B–null minus intact); more negative = more essential in null; red = BH–FDR $q < 0.05$. **(B)** DepMap RNAi (DEMETER2): the same core module silenced orthogonally by shRNA (no DNA cutting), ruling out a copy–number cutting artifact; blue = one–sided $p < 0.05$ (null more dependent). **(C)** GDSC dose–response to the potent FAK inhibitor PF–562271: CDKN2A/B–null lines are selectively more sensitive (lower AUC; Mann–Whitney $p = 2 \times 10^{-3}$). **(D)** CPTAC–GBM phosphoproteomics: FAK autophosphorylation–Y397 (the canonical activation site) is elevated in null tumors ($p = 0.012$)—FAK is more *active*, not just present. **(E)** CPTAC–GBM proteomics: FAK and *FERMT2* protein are both more abundant in null tumors. **(F)** TCGA–GBM RNA–seq: *ITGAV* upregulated in null tumors, with *RB1*/*CDK6*/*E2F1* confirming null–state cell–cycle wiring. Two silencing technologies (A,B), a potent–inhibitor dose–response (C), tumor protein/activation (D,E), and tumor RNA (F) concordantly support the dependency.

significance (§2.1), so we do *not* claim TAZ separates better than FAK—both show comparable, module–level residual dependency once the *MTAP* controls (*PRMT5*/*MAT2A*) are confirmed to collapse. Within null lines TAZ dependency does not differ by *MTAP* status. As for FAK, the separation is statistical and definition–sensitive, not experimental: at $\phi = 0.92$ a joint regression leaves even *PRMT5* unresolved, and isogenic models are required.

Tumor level: abundance, not output activation (honest negative). In CPTAC–GBM (44 deep CDKN2A/B–deleted vs 36 intact tumors), TAZ (*WWTR1*) protein is selectively higher in null tumors and survives adjustment for mesenchymal state (CDKN2A/B coefficient +0.275, $p = 0.007$; *YAP1* unchanged, mirroring the CRISPR paralog specificity), with *TEAD1* protein higher univariately ($p = 0.033$) but not after mesenchymal adjustment. However—stated plainly—the canonical YAP/TAZ–TEAD transcriptional *output* signature (*CTGF*/*CCN2*, *CYR61*/*CCN1*, *ANKRD1*, *AMOTL2*, ...) is *not* elevated in null tumors (RNA $p = 0.79$, protein $p = 0.58$) and is fully explained by mesenchymal state (mesenchymal coefficient $p = 2 \times 10^{-6}$; CDKN2A/B $p = 0.76$; Supplementary Figure S2D). The tumor data thus support elevated *abundance* of the TAZ node, not elevated transcriptional *output*; unlike FAK–Y397, we do not claim output–level activation in tumors.

Why this matters. TAZ/TEAD1 is the transcriptional bottleneck of the same dependency and—unlike the upstream kinase FAK, which has failed as GBM monotherapy—is addressable by clinical–stage TEAD palmitoyl–pocket inhibitors (e.g. VT3989, in a phase 1/2 solid–tumor trial). It is a second, mechanistically distinct handle on the 9p21–codeleted dependency, favouring an axis–level or combination strategy over single–node inhibition. We do not claim TAZ/TEAD supersedes FAK: the FAK module is the more coherent dependency and the only node with tumor activation evidence, whereas TAZ/TEAD is the co–essential, better–separated, and more readily druggable output node.

2.3 The CDK4/6–inhibitor–eligible subset: large, definable, untested—but not prognostic (Figure 2)

CDK4/6 inhibitors are mechanistically rational only when RB1 is intact and the CDK4/6 axis is activated (CDKN2A/B deletion and/or CDK4/CCND amplification) in IDH–wildtype disease. Applying these criteria, ∼244 TCGA and ∼266 MSK–IMPACT IDH–wildtype patients qualify—a large, crisply–definable subset that has not been prospectively enrolled in a CDK4/6–inhibitor trial with resistance gating.

Eligibility, however, did **not** predict better survival (Figure 2A). In TCGA, eligible patients had median OS 12.9 vs 15.3 months (log–rank $p = 0.12$), and in a multivariable Cox model adjusting for age and MGMT methylation the eligibility hazard ratio was 1.32 (95% CI 0.95–1.84, $p = 0.095$)—if anything toward worse outcome. In the independent MSK–IMPACT cohort the comparison was null (21.9 vs 23.6 months, $p = 0.84$). MGMT methylation dominated survival as expected (MSK: 42.2 vs 19.3 months).

Examining the eligible group for known CDK4/6–inhibitor resistance lesions revealed pervasive **PTEN/PI3K–axis escape** (Figure 2B): PTEN mutation in 34% and PTEN homozygous deletion in 12% (overlap unquantified, so the PTEN–altered union is $\leq 46\%$), plus PIK3R1 mutation (11.5%), PIK3CA mutation (10.2%), and NF1 loss (∼10%); classic CCNE1/YAP1/FGFR bypass events were rare ($< 1\%$). This provides a concrete, testable explanation for prior null trials and a design principle: a CDK4/6–inhibitor trial in GBM should exclude or co–target the PTEN/PI3K axis, which contaminates a large fraction of the naively–eligible population.

Summary: this is a patient–selection and trial–design hypothesis with an explicit resistance map—not evidence of efficacy and not a prognostic biomarker.

2.4 A Finnish–founder *MUTYH* allele shows a suggestive germline GBM signal (Supplementary Figure S1)

In the FinnGen R13 loss–of–function gene–burden release (372,626 participants; C3_GBM $N = 467$ cases), the single qualifying *MUTYH* variant was p.Gly155Asp (chr1:45332791 C>T; rs587781864), a ClinVar Pathogenic/Likely–pathogenic allele for *MUTYH*–associated polyposis (VCV000141595) and a Finnish–founder allele enriched 7.3× in gnomAD Finnish versus non–Finnish–European exomes (absent from NFE genomes). It associated with GBM at OR= 24.9 (95% CI 4.3–145.8;

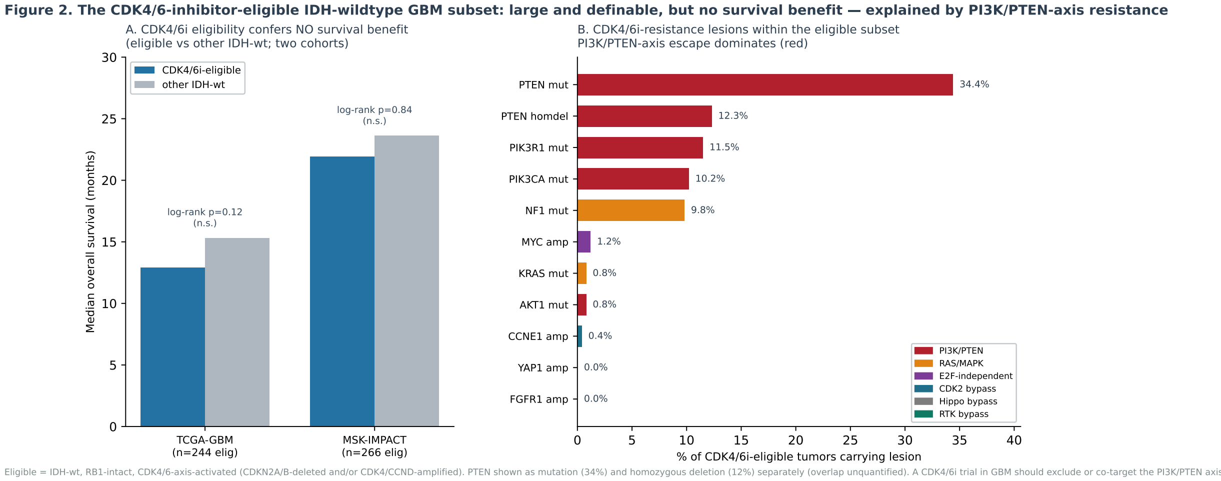

Figure 2: The CDK4/6–inhibitor–eligible IDH–wildtype GBM subset: large and definable, but no survival benefit—explained by PI3K/PTEN–axis resistance. (A) Median overall survival of CDK4/6i–eligible versus other IDH–wildtype patients in TCGA–GBM and MSK–IMPACT; both non–significant ($p = 0.12$, $p = 0.84$). **(B)** CDK4/6i–resistance lesions within the eligible subset (% of eligible tumors), coloured by escape pathway. PTEN mutation (34%) and homozygous deletion (12%) shown separately (overlap unquantified); PI3K/PTEN–axis lesions (red) dominate.

$-\log_{10} P = 3.45$; ∼4 case carriers), with concordant positive direction across GBM, astrocytoma, and brain–wide endpoints.

Two features support plausibility. First, the same allele reproduced the **expected polyposis–associated colorectal/GI phenotypes** (colorectal cancer OR≈3.3, matching published monoallelic estimates), confirming it is functional in FinnGen rather than an annotation artifact. Second, compound heterozygosity was excluded (probability 4.5×10^{-7}), so the GBM signal arises from **monoallelic** carriers—a previously unreported monoallelic–*MUTYH*→GBM hypothesis. The variant lies in the N–terminal adenine–DNA–glycosylase *catalytic* domain (HhH–GPD fold, active site ∼residue 131), consistent with impaired base–excision repair of 8–oxoguanine lesions and a resulting mutator phenotype.

This signal does **not** survive genome–wide multiple–testing correction (Bonferroni threshold for ∼4,789 gene burden tests: $-\log_{10} P > 4.98$) and rests on ∼4 carriers, so the odds ratio is certainly inflated (true OR plausibly 5–10). It is a replication–grade lead—for FinnGen R14+, the Estonian Biobank, or pooled Scandinavian glioma series—not a discovery or clinical recommendation.

2.5 Somatic and germline landscape context

The TCGA–GBM somatic landscape recovered expected driver frequencies (CDKN2A 57%/CDKN2B 56% homozygous deletion, EGFR 53%, PTEN 44%, TP53 33%, CDK4 16% amplification, PDGFRA 16%, NF1 14%, RB1 12%), and survival anchors behaved canonically (IDH1–mutant longer OS, $p = 8 \times 10^{-5}$; TP53–mutant longer OS, $p = 4 \times 10^{-4}$), validating the clinical pipeline used in §2.3.

At the *germline* level, an independent line corroborates the centrality of the CDKN2A/B axis. Of 1,467 GWAS Catalog glioma/glioblastoma associations (168 genome–wide significant), the **CDKN2A/B 9p21 region is a top established germline risk locus** (best $p = 1 \times 10^{-45}$, rs634537; *CDKN2B-AS1* $p = 2 \times 10^{-21}$), alongside the canonical glioma loci *TERT* ($p = 8 \times 10^{-74}$), *RTEL1* ($p = 4 \times 10^{-46}$), *PHLDB1* ($p = 6 \times 10^{-43}$), *TP53* ($p = 9 \times 10^{-38}$) and *EGFR* ($p = 5 \times 10^{-34}$). Inherited variation at CDKN2A/B thus causally raises glioma risk—a germline layer distinct from, but pointing at the same locus as, the somatic–deletion FAK dependency in §2.1. By the same token, *MUTYH* is *absent* from established germline glioma loci, which appropriately tempers the suggestive Finnish–founder signal in §2.4 to a replication–grade lead.

3 Discussion

Across orthogonal datasets, the most robust new signal is a **genetic dependency of CDKN2A/B–null GBM on the integrin/FAK focal–adhesion axis**, now supported by five complementary lines of evidence: genome–wide–significant CRISPR essentiality, orthogonal RNAi silencing (excluding a copy–number cutting artifact), selective sensitivity to the potent FAK inhibitor PF–562271 in dose–response, tumor *ITGAV* mRNA upregulation, and—most functionally—elevated FAK Y397 autophosphorylation and FAK/*FERMT2* protein in CDKN2A/B–null tumors. This fits a model in which loss of the CDKN2A/B locus, beyond releasing the CDK4/6–RB axis, increases reliance on adhesion– and mechanotransduction–driven survival signaling (FAK→Rac/Arp2/3, YAP/TAZ–TEAD). An important constraint, however, is that CDKN2A/B and *MTAP* co–delete on 9p21 and are statistically inseparable in available cell–line panels (§2.1); the dependency is therefore formally a 9p21–codeletion association, with CDKN2A/B the more mechanistically plausible—but not yet experimentally proven—driver. The earlier failure of GSK2256098 in PRISM does not contradict this: it reflects an inert first–generation probe at a single screened dose, and is directly overturned by the PF–562271 dose–response. The actionable next step is a dedicated, potent FAK–inhibitor experiment—alone and combined with CDK4/6 inhibition—in CDKN2A/B–null versus intact GBM models, with clonogenic/3D rather than short–term 2D viability readouts.

A second, druggable node at the output of the axis. The mechanotransduction framing also widens the target options. The terminal effector TAZ/TEAD1 is, by statistical significance, a stronger single–gene dependency than FAK, separates somewhat better from the 9p21/*MTAP* co–deletion, and—crucially—is addressable by clinical–stage TEAD palmitoyl–pocket inhibitors, whereas FAK has failed as a GBM monotherapy (§2.2, Supplementary Figure S2). We are deliberately measured here: the FAK module remains the more coherent dependency and carries the only tumor *activation* evidence (FAK–Y397), while the tumor data for TAZ show elevated node abundance against a mesenchymally–confounded, flat transcriptional output. The constructive reading is therefore not "FAK versus TEAD" but a single integrin→FAK→TAZ/TEAD dependency with two pharmacologically distinct handles—which, together with the CDK4/6–resistance map (§2.3), argues for biomarker–selected *combination* targeting of the 9p21–null subtype rather than any single agent.

Relation to prior FAK–inhibitor experience, and why this is a hypothesis rather than a repurposing recommendation. FAK has been pursued as an oncology target for over a decade, and the relevant clinical history is largely negative, which we state plainly. FAK inhibitors (e.g. defactinib) have shown little single–agent activity in unselected solid tumors, and FAK–inhibitor trials in glioblastoma have not demonstrated benefit. Most directly, the one biomarker–stratified test of a CDKN2A–FAK hypothesis—a phase 2 study of defactinib in *KRAS*–mutant NSCLC that prospectively assigned patients by CDKN2A and TP53 status—was negative, with efficacy explicitly *not* correlated with CDKN2A status (Gerber et al., 2020). Two features distinguish our hypothesis from that precedent rather than excuse it. First, the NSCLC synthetic lethality was *KRAS–conditional* (FAK dependence required mutant *KRAS* together with CDKN2A/TP53 loss); IDH–wildtype GBM is not *KRAS*–driven, so the proposed dependency, if real, would operate through a different mechanism (CDK4/6–RB rewiring and adhesion/mechanotransduction) and cannot be assumed to share the NSCLC trial's outcome. Second, our evidence is dependency– and activation–level (orthogonal CRISPR/RNAi essentiality plus elevated FAK–Y397 in tumors), which the prior trial did not use for selection. Nonetheless this history sets a high bar: it means a positive isogenic and CNS–orthotopic result—ideally with a defined combination partner (CDK4/6 or PI3K–axis inhibition) and a 9p21–null selection marker—is a prerequisite before any clinical inference, and that monotherapy in unselected GBM is expected to fail. We therefore frame finding (1) as a novel, multi–evidence *dependency hypothesis* for the 9p21–codeleted subtype, not as a ready–to–repurpose therapy.

The CDK4/6–inhibitor analysis reframes a recurring clinical disappointment: the eligible subset is real and large, but heavily contaminated by PTEN/PI3K–axis lesions that bypass CDK4/6 blockade. A trial that selects RB1–intact, CDK4/6–axis–on tumors *and* excludes or co–targets the PI3K axis is the logical, never–executed experiment. The *MUTYH* germline observation is the narrowest finding but is biologically coherent and exploits a genuinely Finnish–private founder allele; it warrants targeted replication rather than over–interpretation.

4 Limitations

- **FAK pharmacology still needs wet–lab confirmation.** Five concordant lines (CRISPR, RNAi, GDSC PF–562271, TCGA RNA, CPTAC protein/phospho) establish a genetic and protein–level dependency, but all are computational/associative; we performed no wet–lab validation. The GDSC drug effect, though significant, is modest ($\sim$2% AUC shift), the CNS–restricted subsets are underpowered (direction–consistent, often non–significant), and CPTAC tumor associations are correlative, not causal. *VCL* did not replicate under RNAi, and *ITGAV* protein was not elevated despite higher mRNA—the protein/activation signal is specifically FAK+*FERMT2*.
- **CDKN2A/B and *MTAP* are only partially separable.** The two genes co–delete on 9p21 in 92% of null lines ($\phi = 0.92$), so a joint regression cannot resolve even the positive control (the known *MTAP* dependency *PRMT5* misattributes to CDKN2A/B). A subset contrast (*MTAP*–intact null vs intact) with a permutation/bootstrap meta–analysis does dissociate them—the *MTAP*–loss controls *PRMT5*/*MAT2A* collapse while the focal–adhesion module persists (pooled *PTK2* $p = 0.017$, *FERMT2* $p = 0.014$, *PXN* $p = 0.038$)—but no single gene survives multiple–testing, the effect weakens under a strict homozygous–both deletion call ($n = 6$) and, for the TAZ pan–cancer contrast, under permutation ($p = 0.068$). The dependency is therefore best described as a 9p21–codeletion association with CDKN2A/B the more likely driver; definitive attribution requires isogenic CDKN2A/B restoration with *MTAP* held constant.
- **The YAP/TAZ–TEAD output node is abundance– not activation–validated in tumors.** *WWTR1*/TAZ and *TEAD1* are selectively essential and TAZ protein is elevated in null tumors (mesenchyme–adjusted $p = 0.007$), but the canonical YAP/TAZ transcriptional *output* signature is not elevated and is fully explained by mesenchymal state ($p = 2 \times 10^{-6}$), and *TEAD1* protein elevation does not survive mesenchymal adjustment—so we do not claim output–level pathway activation in tumors. Like FAK, the dependency is a 9p21–codeletion association only partially separable from *MTAP* (§2.2); and paralog redundancy (*YAP1*$\leftrightarrow$*WWTR1*, *TEAD1–4*) means single–gene CRISPR understates pathway dependence and that only a pan–TEAD agent would test it.

- **CDK4/6–eligibility shows no survival benefit** and must not be read as efficacy; the cohorts are retrospective and treatment–heterogeneous.
- ***MUTYH*** rests on ∼4 carriers, fails Bonferroni correction, and is from a single biobank; the odds ratio is inflated.
- Cell–line and TCGA analyses use copy–number thresholds that approximate but do not perfectly capture homozygous deletion; cohort molecular annotations are imperfect.
- This is a hypothesis–generating computational study; none of the findings constitute clinical guidance.

5 Methods (summary)

Cell–line dependency (CRISPR). DepMap CRISPRGeneEffect and OmicsCNGene (24Q4); CNS lineage by OncotreeLineage; CDKN2A/B–null = relative CN<0.2 in either gene; Mann–Whitney U per gene (null vs intact), Benjamini–Hochberg FDR. **Orthogonal RNAi.** DepMap RNAi DEMETER2 combined gene–dependency scores; cell lines mapped to CN status via Model.csv (CCLEName); per–gene Mann–Whitney (one– and two–sided, null vs intact), pan–cancer and CNS subset. **Tumor expression.** TCGA–GBM RNA–seq via cBioPortal; same CN classification; per–gene Mann–Whitney, BH–FDR. **Drug response.** PRISM Repurposing LFC_COLLAPSED (GSK2256098); and GDSC2 fitted dose–response (release 8.5) for PF–562271 (DRUG_ID 158), cell lines classified by DepMap CN via SANGER_MODEL_ID/COSMIC_ID; Mann–Whitney on AUC and $\ln IC_{50}$ (null vs intact). **Tumor proteome/phospho.** CPTAC–GBM (`cptac` package): CNV (`bcm`, log2 gene ratio), proteomics and phosphoproteomics (`umich`); tumors classified deep–loss (log2≤ -1.0 both genes) vs intact (>-0.3 both), ambiguous excluded; Mann–Whitney null vs intact on FAK Y397 phospho, FAK/*FERMT2* protein. **Mesenchymal–confound control.** CPTAC–GBM transcriptomics (`washu`); mesenchymal and FAK–module signature scores as mean of z–scored genes; Spearman correlation; OLS `FAK ~ MES + is_null`. **Single–cell mesenchymal deconvolution.** Neftel et al. 2019 GBM atlas (GSE131928): Smart–seq2 (GSM3828672, processed $\log_2$(TPM/10+1)) and 10x (GSM3828673, TPM) IDH–wildtype cohorts, patient identity from cell barcodes. Malignant cells retained by signature filtering—tumor–state score (max of MES/AC/OPC/NPC) exceeding the immune/oligodendrocyte/endothelial/fibroblast–pericyte maximum, immune and stromal scores <0.10—excluding non–malignant mesenchymal stroma. Per–cell scanpy `score_genes` for the four Neftel states, the FAK module and a YAP/TAZ–output signature; cells assigned to the argmax state. Pseudoreplication controlled by patient–level inference: pseudobulk mean scores per patient×state, patient–level Pearson correlation of mean FAK vs mean MES ($n = 31$–33 patients with ≥ 20 malignant cells), a linear mixed model `FAK ~ MES + (1|patient)`, and a pseudobulk test of residual FAK across states. ***MTAP* co–deletion control.** CNS and pan–cancer DepMap lines stratified by CDKN2A/B–null × *MTAP*–null (CN<0.2); FAK–module differential essentiality compared across strata (Mann–Whitney) and by OLS effect ∼ `cdkn_null` + `mtap_null`; *PRMT5*/*MAT2A* as positive control for the *MTAP*–loss dependency. **Permutation/bootstrap *MTAP*–separation meta–analysis.** For the deconfounded contrast (*MTAP*–intact null, *MTAP* CN≥0.2, vs 9p21–intact), per–gene effect = median(intact)−median(null) on CRISPR (Chronos) and RNAi (DEMETER2) scores; significance by 50,000–label permutation and a 20,000–resample bootstrap 95% CI; CRISPR and RNAi combined by Stouffer's Z weighted by $\sqrt{n_1 + n_2}$; run under both the canonical (CN<0.2 in either gene) and a strict homozygous (both genes <0.2) deletion call as sensitivity; *PRMT5*/*MAT2A* as *MTAP*–loss positive controls. **YAP/TAZ–TEAD output node.** The same genome–wide CNS CRISPR scan read out for YAP/TAZ–TEAD effectors (*YAP1*, *WWTR1*, *TEAD1–4*) and, as an orientation control, the Hippo negative regulators (*NF2*, *LATS1/2*, *STK3/4*, *SAV1*, *MOB1A/B*); *MTAP*–separation as above. Tumor validation in CPTAC–GBM transcriptomics (`washu`) and proteomics (`umich`): a canonical TEAD–target output signature (*CTGF/CCN2*, *CYR61/CCN1*, *ANKRD1*, *AMOTL2*, *NUAK2*, *CRIM1*, *F3*, *TGFB2*) and components *WWTR1/YAP1/TEAD1* as mean z–scores, null vs intact (Mann–Whitney), with OLS adjustment for a non–overlapping mesenchymal score (*CHI3L1*, *CD44*, *VIM*, *MET*, *RELB*, *SERPINE1*). **Somatic/clinical.** TCGA–GBM and MSK–IMPACT via cBioPortal; Kaplan–Meier/log–rank and Cox proportional hazards (age, MGMT). **Germline.** FinnGen R13 LoF gene–burden table (REGENIE–style, functional mask AF<0.01); annotation via Ensembl VEP and ClinVar; allele frequencies from gnomAD v4; compound–het probability from population AF. **Germline glioma loci.** GWAS Catalog associations for glioma and glioblastoma (REST `findByEfoTrait`); genome–wide significance $p \leq 5 \times 10^{-8}$.

6 Data availability

All primary datasets are public: DepMap CRISPR and RNAi/DEMETER2 (depmap.org), PRISM (depmap.org/repurposing), GDSC release 8.5 (cancerrxgene.org), CPTAC–GBM (`cptac` Python package / proteomic.datacommons.cancer.gov), TCGA–GBM and MSK–IMPACT (cBioPortal), GWAS Catalog (ebi.ac.uk/gwas), FinnGen R13 (finngen.fi), gnomAD v4 (gnomad.broadinstitute.org), ClinVar. Analysis scripts and derived summary tables are available at `https://github.com/bradleymoore1/gbm-cdkn2` (FinnGen individual–level data excepted, per FinnGen's data–access terms).

Acknowledgments

This work was conducted in memory of **James E. Moore** (July 4, 1961 – August 14, 2025), who was diagnosed with IDH–wildtype, MGMT–unmethylated, CDKN2A/B–homozygously–deleted glioblastoma and died of this disease. The analysis was motivated by his illness.

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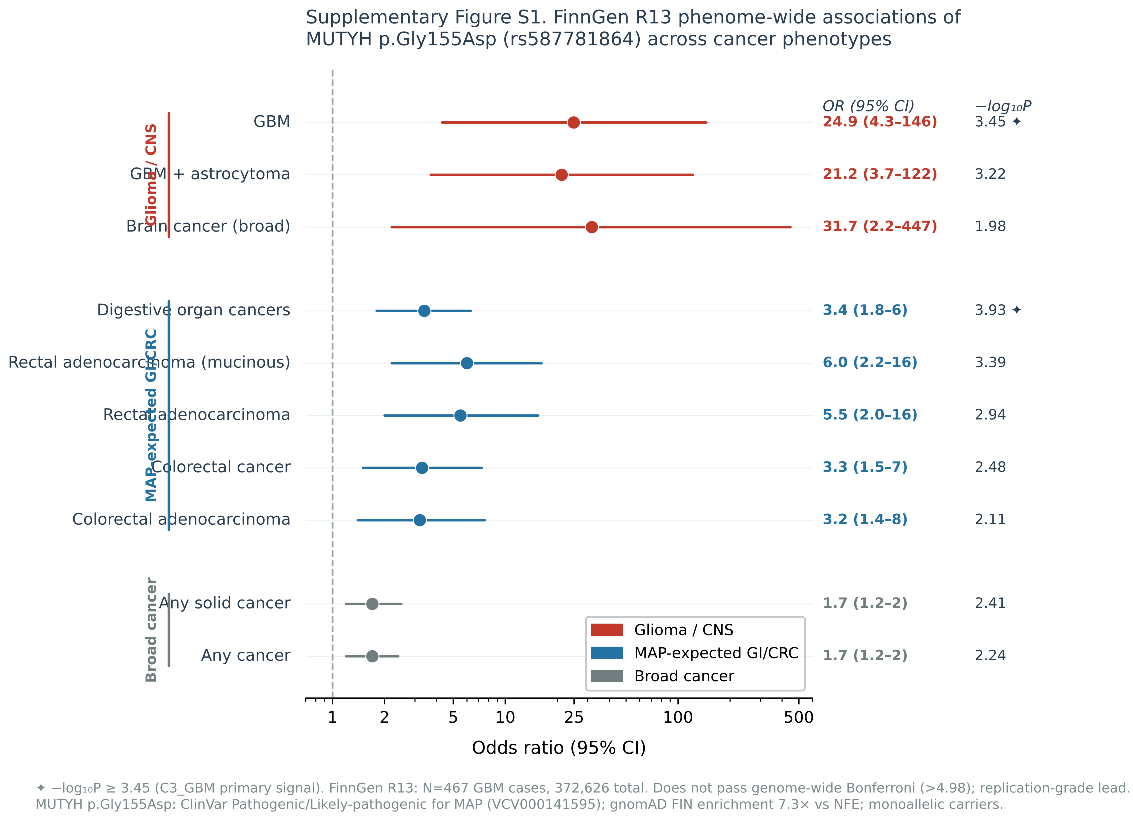
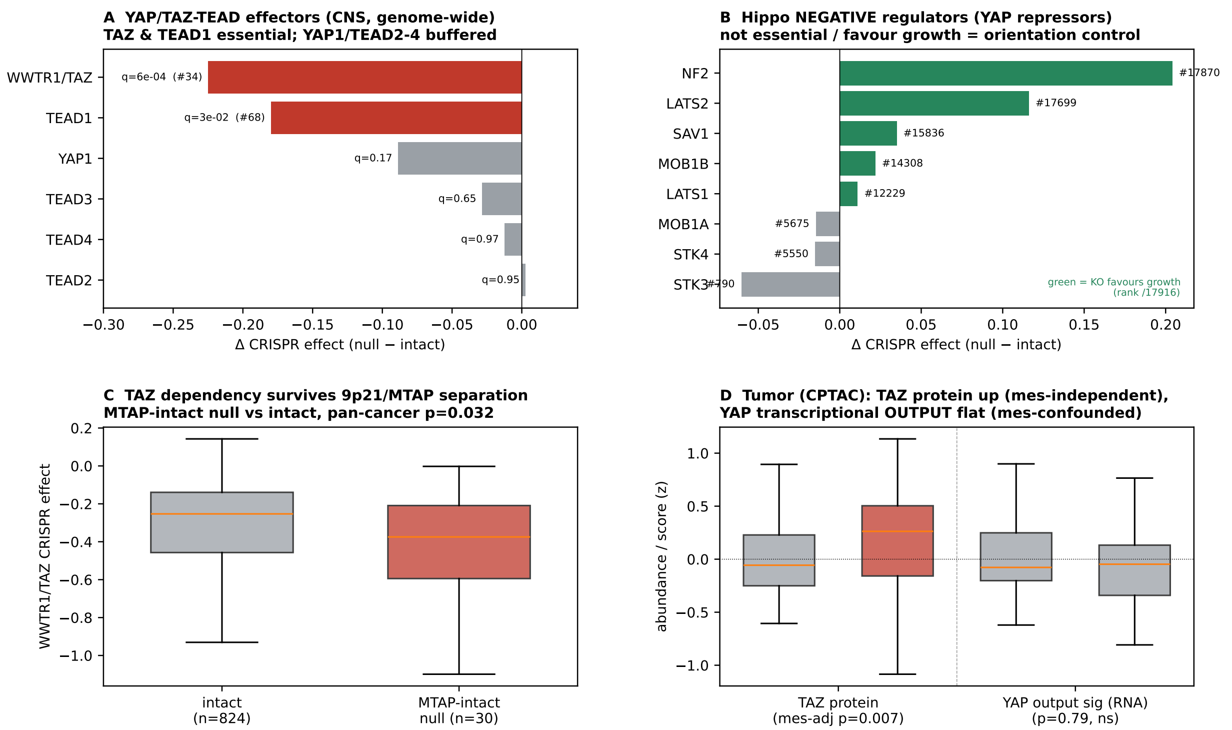
Supplementary Figure S1.



FinnGen R13 phenome–wide associations of *MUTYH* p.Gly155Asp (rs587781864) across cancer phenotypes. Odds ratios (95% CI, log scale) for glioma/CNS (red), polyposis–expected GI/CRC (blue), and broad cancer (grey) endpoints, with $-\log_{10} P$ annotated. The GBM signal (OR$=$ 24.9) does not pass genome–wide Bonferroni correction; the concordant colorectal/GI signals internally validate the allele as functional.

Supplementary Figure S2.

Supplementary Figure S2 | TAZ/TEAD1: the co-essential, MTAP-separable, druggable OUTPUT node of the adhesion axis



The transcriptional–output node of the adhesion axis in CDKN2A/B–null GBM. **(A)** Genome–wide CNS CRISPR differential essentiality for YAP/TAZ–TEAD effectors: *WWTR1*/TAZ and *TEAD1* are selectively essential (red, BH $q < 0.05$; genome–wide rank annotated), whereas *YAP1* and *TEAD2/3/4* are buffered by paralog redundancy. **(B)** Hippo *negative* regulators (YAP/TAZ repressors) are not essential and several favour growth when knocked out (green; rank /17,916), the orientation control for a YAP/TAZ–dependent state. **(C)** TAZ is more essential in *MTAP*–intact null lines than in intact lines (pan–cancer Mann–Whitney $p = 0.032$; the more conservative pooled CRISPR+RNAi permutation gives $p = 0.068$), partially separating the dependency from the 9p21/*MTAP* co–deletion. **(D)** CPTAC–GBM: TAZ protein is selectively higher in null tumors after adjustment for mesenchymal state ($p = 0.007$), but the canonical YAP/TAZ transcriptional output signature is flat ($p = 0.79$, mesenchymally confounded)—abundance of the node, not output activation.